

1. Trig Review

We can't talk about the physics of music without using sines and cosines. They are fundamental to everything we're going to do mathematically. Your ear and brain also work on sines/cosines - you perceive a sine wave as a pure tone. Because they're so fundamental, we're going to review them here.

You probably learned in trigonometry class that in a right triangle, the sine of an angle is the length of the opposite side divided by the length of the hypotenuse. The cosine is the length of the adjacent side divided by the hypotenuse. If you look at Figure 1, then for angle θ , we have $\cos(\theta) = a/c$ and $\sin(\theta) = b/c$.

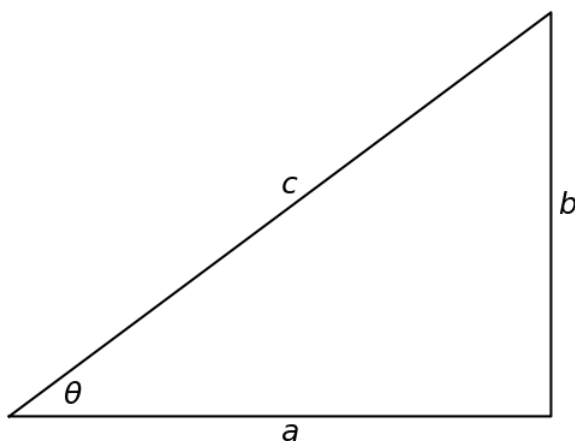


Fig. 1.— Plot of a right triangle. If a, b, c are the length of the corresponding sides in the figure, then we have $\cos(\theta) = a/c$ and $\sin(\theta) = b/c$.

Before moving on, we need to talk about the units we use for measuring angles. In ordinary, everyday life, we usually use degrees. There are 360 degrees in a circle, 90 degrees in a right angle, and so on. In the right triangle in Figure 1, θ must be somewhere between 0 and 90 degrees. However, nature has decided that the natural unit for measuring angles is the *radian*, where there are 2π radians in a circle. A right angle is a quarter of a circle, or $2\pi/4 = \pi/2$ radians. One radian is equivalent to $360/2\pi = 180/\pi = 57.295\dots$ degrees. If we use radians, then the sine of a small angle is approximately equal to the angle

$$\sin(\theta) \equiv \theta \tag{1}$$

This simple (and very, very useful!) approximation is only true if we measure our angles in radians. Similarly, the cosine of a small angle is

$$\cos(\theta) \equiv 1 - \frac{\theta^2}{2} \tag{2}$$

Again, this is only true if we use radians. Euler's formula is one of the most beautiful results in all of mathematics, linking exponentials, imaginary numbers, and trig functions:

$$e^{i\theta} = \cos(\theta) + i\sin(\theta) \quad (3)$$

where $e = \lim_{n \rightarrow \infty} (1 + 1/n)^n \equiv 2.182818459045\dots$ is the natural base for exponentials and logarithms, and i is $\sqrt{-1}$. Once again, this is only true if we are using radians¹. Don't worry if you don't have the math background to follow Euler's formula - I'm only citing it here as yet another reason that radians are the natural way to measure angles. Unless otherwise specified, any angle in this class will be assumed to be in radians.

With the right triangle definition, sine and cosine are only defined between 0 and 90 degrees, or since we're using radians, we have $0 < \theta < \pi/2$. However, right triangles turn out not to be a very good way of defining sine/cosine because of this. A better definition is to start at the origin in the (x, y) plane, and draw a line of length 1 at an angle θ from the x -axis. The (x, y) coordinates of the end of the line are then $\cos(\theta)$ and $\sin(\theta)$. Then θ can be any value, positive or negative, and sine/cosine are well-defined. Since we wrap around a full circle after 2π radians, then it should be clear that $\cos(\theta + 2\pi) = \cos(\theta)$ for *any* value of θ , and similarly with $\sin$.

To visualize this, we'll look at some plots using desmos.com. Desmos will animate various functions for you. We can see the circle version of sine/cosine by going to [this desmos link](#). If you watch the red line that has length 1 as it rotates around a circle, the height of the tip is the sine of the angle between the line and the x -axis, and the horizontal position of the tip is the cosine. Obviously, sine and cosine repeat as the line goes around in a full circle. Since the line has length 1, the absolute values of sine and cosine are always less than or equal to 1, but they can be positive or negative. If I were to take my circle and rotate it by 90 degrees (or $\pi/2$ radians!), then nothing qualitatively changes, but I've now swapped the x - and y -axes, which means I've swapped sine and cosine. They're really the same function, just shifted over from each other.

This shift is clear if instead of watching a circle rotate, we change our plot to have the angle along the x -axis, and have sine and cosine plotted separately. You can see [this here](#). The animation shows sine and cosine as the angle goes around the circle roughly three times (or $6\pi \sim 19$ radians). At this point, I hope you can start to see that sine and cosine are really the same function, just shifted relative to each other. If I take a cosine and shift to the right by $\pi/2$ radians, then I get a sine. You look at [yet another desmos animation](#) that keeps sine fixed while shifting cosine. If you pause the animation right when cosine overlaps with sine, then the shift (given by the value of t) will be $\pi/2 = 1.57\dots$

We'll often be faced in this course with needing a half period of a sine/cosine wave. We could say pick our x -range to go from 0 to π , in which case $\sin(x)$ will start at 0, go up to 1 in the middle of our x -range, and go back to 0 at the end. Similarly, if we instead decided our x -range was $-\pi/2$ to $\pi/2$, then $\cos(x)$ would behave exactly in the same way. There's nothing particularly special about either option and they're both perfectly correct to use. As long as you're clear about your choice, we're happy to accept any correct answer.

¹More explicitly, if we pick a value for θ , then $\exp(i\theta)$ is well defined. At this point θ is just a number, but the value for $\exp(i\theta)$ is the sine/cosine of θ radians.

2. Sums and Differences

How the sines and cosines of sums and differences of angles work is something that will show up over and over again. It's important in understanding what's going on with a lot of acoustic phenomena, including why out-of-tune instruments sound so, so bad.

Given two angles α and β , we want to know the sine and cosine of $\alpha + \beta$. There are many, many ways to derive the answers. Rather than just quote the result, I'll show you a derivation that uses Euler's formula. It's OK if you don't follow this proof - I don't expect you to be experts with complex numbers. You can prove the same results with traditional trigonometry, but it's a lot more painful. First, recall that

$$c^a c^b = c^{a+b} \quad (4)$$

for *any* values of a, b, c . In particular, we can set $c = e$, $a = i\alpha$ and $b = i\beta$ where i is the square root of -1. With this substitution, we have

$$e^{i(\alpha+\beta)} = e^{i\alpha} e^{i\beta} \quad (5)$$

If we use Euler's formula $e^{ix} = \cos(x) + i \sin(x)$ we can expand Equation 5 to get:

$$\cos(\alpha + \beta) + i \sin(\alpha + \beta) = (\cos(\alpha) + i \sin(\alpha)) (\cos(\beta) + i \sin(\beta)) \quad (6)$$

If we multiply out the right hand side (and remember that $i * i = -1$) we get:

$$\cos(\alpha + \beta) + i \sin(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) + i (\cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta)) \quad (7)$$

For Equation 7 to hold, then both the real (no i) and imaginary (with an i) parts need to be equal. We can just read those off to get:

$$\cos(\alpha + \beta) = \cos(\alpha) \cos(\beta) - \sin(\alpha) \sin(\beta) \quad (8)$$

$$\sin(\alpha + \beta) = \cos(\alpha) \sin(\beta) + \sin(\alpha) \cos(\beta) \quad (9)$$

I won't hold you responsible for this (or any!) derivation of the sum formulas, but you will need to be able to use Equations 9. That is about as hard as the math you'll be expected to use will get in this course, though.

We can also use Equations 9 to work out the sin/cosine of the *difference* of angles. Remember that $\cos(-x) = \cos(x)$ and $\sin(-x) = -\sin(x)$. If we want to know $\sin(\alpha - \beta)$, we can rewrite this as $\sin(\alpha + (-\beta))$ (similar for cosine) and plug into Equations 9. For cosine, we have

$$\cos(\alpha - \beta) = \cos(\alpha) \cos(-\beta) - \sin(\alpha) \sin(-\beta) = \cos(\alpha) \cos(\beta) + \sin(\alpha) \sin(\beta) \quad (10)$$

For sine, we have

$$\sin(\alpha - \beta) = \cos(\alpha) \sin(-\beta) + \sin(\alpha) \cos(-\beta) = \sin(\alpha) \cos(\beta) - \cos(\alpha) \sin(\beta) \quad (11)$$

3. Beating

Now we have all the tools we need to look at the phenomenon of *beating*. It turns out that if we have a pure tone, we can write the pressure P our ear feels as follows:

$$P = \cos(2\pi\nu t) \tag{12}$$

where ν is the frequency of the sound (for instance, 440 Hz for the A above middle C), and t is the time. Two aspiring violinists *try* to play the same note, but one is a little sharp, the other is a little flat. If both violinists are off their target frequency by an amount δ (which we hope is small!), then the pressure our ear feels is the sum of the two signals:

$$P = \cos 2\pi(\nu + \delta)t + \cos 2\pi(\nu - \delta)t \tag{13}$$

We can expand using the sum/difference formulas:

$$P = \cos 2\pi\nu t \cos 2\pi\delta t - \sin 2\pi\nu t \sin 2\pi\delta t + \cos 2\pi\nu t \cos 2\pi\delta t + \sin 2\pi\nu t \sin 2\pi\delta t \tag{14}$$

The terms with cosine are the same, so they add. The terms with the sines look the same, except one is positive and one is negative. When we add them together, they cancel. That leaves us with:

$$P = 2 \cos 2\pi\nu t \cos 2\pi\delta t \tag{15}$$

This result is *true* for any values of ν and δ . It is *interesting* (at least for our purposes) for values where δ is much less than ν .

First, note that over a fixed range of t , the larger ν is, the more oscillations of $\cos 2\pi\nu t$ you get. Say that t goes from 0 to 1, and $\nu = 1$. In that case, the inside argument to $\cos$ goes from 0 to 2π , or one complete cycle of a sine wave. If we make $\nu = 10$, say, then the inside argument goes from 0 to 20π , or 10 complete cycles of a sine wave. The larger ν gets, the more cycles we get. This would be a great thing to practice looking at using desmos! If δ is much smaller than ν , then we can think about this as a quickly-oscillating sine wave driven by ν that's multiplied by an overall envelope, driven by the δ term. You can see this behavior with some sensible defaults [here](#).

Essentially, what's happening is that the two signals start out in-phase with each other, so the sum of the two sine waves is a bigger sine wave. As time goes on, though, the sine waves get out of alignment, and they cancel each other out. Wait even longer, though, and they get back in phase and reinforce each other again. If you've ever heard an unpleasant "wah wah wah wah" sound from out-of-tune instruments, this is exactly what's going on. Sometimes their sounds reinforce each other, sometimes they cancel each other out. What you hear is a tone that gets louder and softer with time. I've included an audio example on github ([beating.300.5.wav](#)) so you can hear what happens when you have 2 300 Hz signals that are 5 Hz apart. What you hear is a 300 Hz sound (about the D above middle C) that gets louder and softer 5 times per second. The warbling sound is known as "beating", and can be used to tune instruments. If one instrument is defined to be in-tune, then other instruments should be tuned to make the beating go away when you play at the same time as the in-tune one. Piano tuners also use this to tune different keys - they try to make the beating go away as much as possible².

²Because of the physics of pianos, you can't actually do this exactly. Tuners do the best job they can given the

4. Summary and Discussion

We've reviewed some of the properties of sine and cosine here. These are the fundamental functions that we'll use to describe musical sounds, so it's important to get familiar with them/refresh your memory. We also looked at the sine/cosine of the sum/difference of angles. We derived that using complex numbers, but don't worry if you didn't follow it; you won't be responsible for complex math. You will however need to be able to use the formulas, as they show up over and over again. Finally, we showed how we can use the sum/difference formulas in a practical application. When two signals are almost, but not quite, in tune, we hear the total sound warble as it gets louder/softer with time. This phenomenon is known as "beating", and we'll look at it in more detail later.

realities of the pianos. We'll hopefully look more at this towards the end of the course!